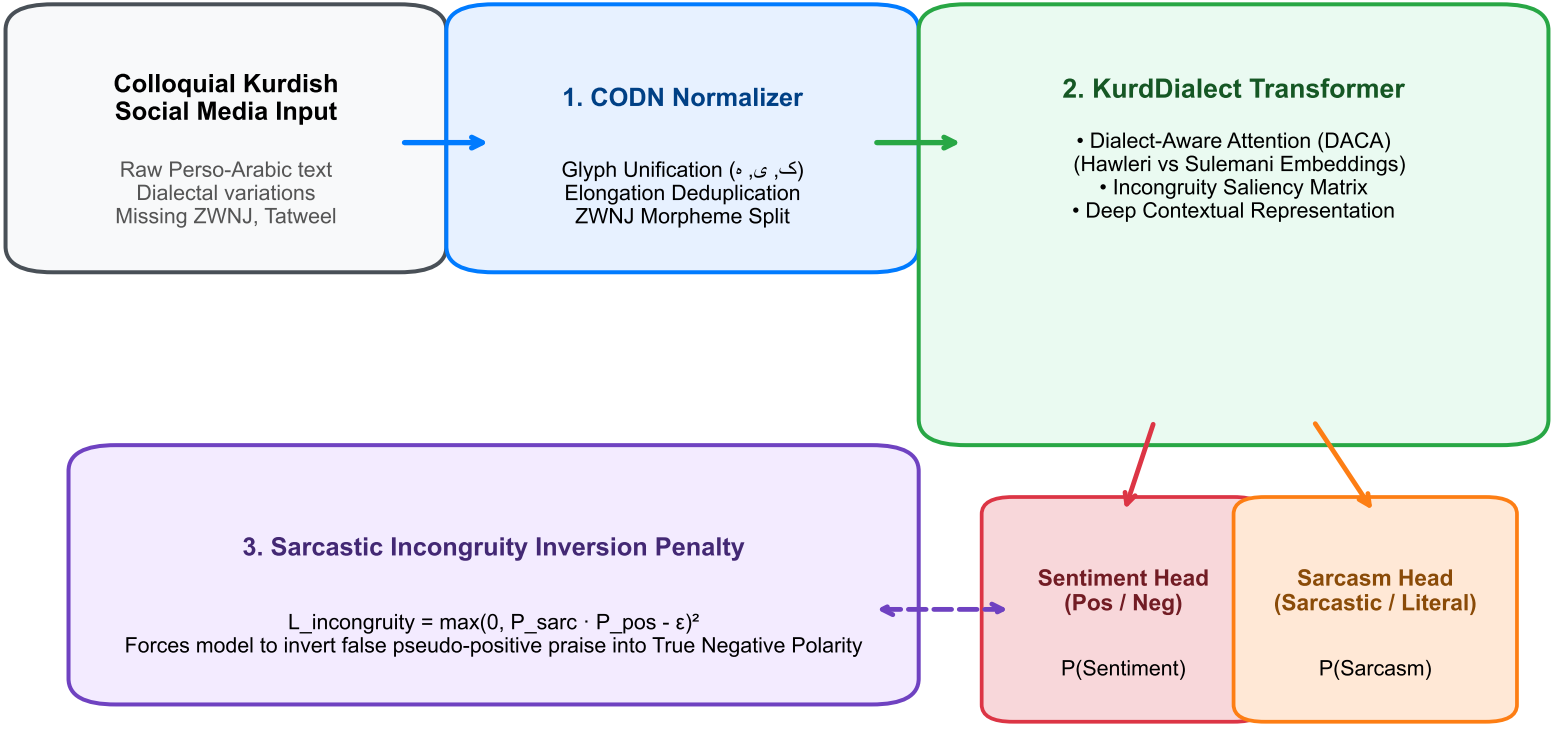
Colloquial Kurdish
Social Media Input
Raw Perso-Arabic text
Dialectal variations
Missing ZWNJ, Tatweel
1. CODN Normalizer
Glyph Unification (ک, ی, ه)
Elongation Deduplication
ZWNJ Morpheme Split
2. KurdDialect Transformer
• Dialect-Aware Attention (DACA)
(Hawleri vs Sulemani Embeddings)
• Incongruity Saliency Matrix
• Deep Contextual Representation
3. Sarcastic Incongruity Inversion Penalty
L_incongruity = max(0, P_sarc · P_pos - ε)²
Forces model to invert false pseudo-positive praise into True Negative Polarity
Sentiment Head
(Pos / Neg)
P(Sentiment)
Sarcasm Head
(Sarcastic / Literal)
P(Sarcasm)